The Purpose of a Character: Identity as a Conserved Invariant and Purpose as a Fixed Point, for a Single Agent and for Agents in Society

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Abstract

We ask what it is for a synthetic character—an agent in a simulated world—to *have* an identity and a *purpose*, and we answer the question with mathematics that a reader can follow without reference to any other work. We model an agent as a finite weighted graph, the *self-graph*, whose vertices are the distinctions the agent draws within itself and whose edge weights are the costs of holding those distinctions apart. From three axioms—the agent is finite, it is never in a null state, and no self-distinction is free—we prove a positive *floor* on every internal separation and derive two objects that together constitute the agent's character. **(T1)** Each agent carries a *conserved invariant*, computed from its own internal boundary content, that is preserved under every relabelling of its parts and is realised by no single part; this invariant is the agent's *identity*, the standing structure from which its changing behaviour derives, and it is a region, never a point. **(T2)** An agent's moment-to-moment conduct is a walk on its self-graph, undetermined by the graph alone; selecting the next act requires an additional datum, a *drive field*, and the field's stable set—its fixed point—is the agent's *purpose*: the region of behaviour the agent is drawn to and returns toward. **(T3)** Purpose exists and is unique up to an equivalence we make precise; **(T4)** it is an attractor, so conduct perturbed away from purpose returns to it, giving characters that are robust rather than brittle. We then place the agent in *society*. **(T5)** A collection of agents acting as one admits a single shared drive on the quotient of their self-graphs, and exactly one; two drives are the signature of a group that is really two. **(T6)** Coordinated purpose among agents requires no shared identity, shared beliefs, or shared internal vocabulary: agents with wholly disjoint self-graphs can share a purpose defined in the common space of outcomes, and **(T7)** a crowd of such agents settles on a purpose more sharply than any one of them. Finally **(T8)** an agent's identity is unreachable from outside: no finite interrogation resolves it to a point, and an authored copy of a character is always a distinct individual with its own history. We give constructions, an attractor analysis, and a numerical study on explicitly built agents. The interpretive readings—identity as character, drive as motivation, purpose as goal, the quotient as a collective—are marked as such and no theorem depends on them.

Keywords: agent identity; purpose; conserved invariant; gated walk; fixed point; attractor; quotient graph; coordination; non-player character.

1 Introduction

1.1 The question

A character in a simulated world is convincing when it seems to have a life of its own: a standing identity that persists across encounters, and a purpose it pursues whether or not anyone is watching. The difficulty is not to make an agent talk or react—that is comparatively easy—but to make it *present*: to give it an interior it is not always turning outward. A merely reactive agent has no identity of its own; it is a mirror of whoever addresses it. What distinguishes a character from a reaction is that the character has something conserved inside it that its behaviour derives from, and something it is trying to do that is its own. The question is a foundational one for synthetic agents and non-player characters [17, 20].

This paper asks what those two things *are*, formally. We want a definition of a character's *identity* and of its *purpose* that is precise enough to compute with, general enough to apply to any bounded agent, and self-contained: the reader should need nothing but this paper and standard mathematics.

1.2 Method

We prove a conditional. *If* an agent is a finite thing that is always in some state and that cannot draw an internal distinction for free, *then* it necessarily has an identity of a determinate kind and admits a purpose of a determinate kind. We assume nothing about what the agent is made of. The deductions are combinatorial (finite graphs and their symmetries), order-theoretic (walks and their histories), and elementary in dynamics (fixed points and attractors of a gradient flow). Where a tempting claim could not be proved at this standard we have removed it.

Our strategy throughout is to model the agent as a finite weighted graph and to read identity, behaviour, and purpose off its structure:

- the *self-graph* is the agent's internal structure—the distinctions it holds within itself and the cost of holding them;
- *identity* is the invariant of that graph that survives all re-description;
- *behaviour* is a walk on the graph;
- *purpose* is the fixed point of the field that selects the walk.

1.3 What we do and do not claim

We do not claim that any particular agent architecture is correct, nor that biological or human identity is a finite graph. We claim that *any* bounded agent satisfying the three axioms has the structure we derive, and that this structure is enough to found a usable notion of character and purpose for synthetic agents. The reading of the formal objects as "character," "motivation," and "goal" is offered as orientation and is confined to remarks; the theorems are graph- and order-theoretic and are used only as such.

1.4 Organisation

Section 2 fixes the agent as a self-graph and states the axioms. Section 3 proves the floor. Section 4 derives identity (T1). Section 5 makes behaviour a gated walk and proves that selection needs a drive (T2). Section 6 defines purpose as a fixed point and proves existence, uniqueness, and attraction (T3, T4). Section 7 places agents in society and proves the single-drive, disjoint-coordination, and crowd-sharpening results (T5–T7). Section 8 proves identity is unreachable from outside and copies are distinct (T8). Section 9 reports a numerical study. Section 10 discusses scope.

2 The agent as a self-graph

We build the agent from the inside. An agent is not a point; it is a system of internal distinctions—this mood as against that, this commitment as against its abandonment, this memory as against its absence. We record those distinctions as a graph, using standard notions of finite weighted graphs and their cuts throughout [4, 8, 22].

Definition 2.1 (Self-graph)**.** The *self-graph* of an agent A is a finite weighted graph $\Gamma = (V, E, w)$ where V is a finite set of *parts* (the internal distinctions the agent draws within itself), $E \subseteq \binom{V}{2}$ is a set of *separations* (pairs of parts the agent holds apart), and $w : E \to (0, \infty)$ assigns to each separation the *cost* of holding it. The graph is simple and, by Axiom 2.4 below, connected.

Definition 2.2 (Cut and boundary)**.** For disjoint $S, T \subseteq V$, the *cut* between them is $\partial(S,T) = \{uv \in E : u \in S, v \in T\}$ with weight $w(\partial(S,T)) = \sum_{e\in\partial(S,T)} w(e)$. For $\varnothing \neq U \subsetneq V$ the *boundary* of U is $\partial(U) = \partial(U, V \setminus U)$ and its *boundary cost* is $b(U) = w(\partial(U))$.

We now state the three axioms. They are meant to be almost undeniable for anything we would call a bounded agent.

Axiom 2.3 (Finiteness)**.** The self-graph is finite: $|V| < \infty$. An agent holds finitely many internal distinctions at a time.

Axiom 2.4 (Presence, or no null state)**.** At every instant the agent occupies exactly one part; there is no null part and no gap between parts. Equivalently, the self-graph is connected and nonempty: the agent is always somewhere in itself.

Axiom 2.5 (Costly individuation)**.** No internal distinction is free: every separation has positive cost, and these costs are bounded below by a constant. Formally, there is $\beta > 0$ with $w(e) \geq \beta$ for all $e \in E$.

Remark 2.6 (Why costly individuation is not an extra assumption)**.** Axiom 2.5 can be motivated rather than merely posited. To individuate a part U of the agent is to tell it apart from the rest of the agent, $V \setminus U$. If the agent is a genuine whole—if distinguishing a part against the rest were a task that could be *finished*, the rest would be exhausted and the agent would be merely that finite catalogue with nothing held in reserve. An agent that is always in some state (Axiom 2.4) and never reduces to a closed catalogue leaves, on each of its parts, an irreducible remainder of the comparison against the rest; that remainder is the positive floor. We do not rely on this motivation—we take Axiom 2.5 as an axiom and prove everything from it—but it explains why the floor is the natural, not an arbitrary, hypothesis. The floor is the trace the agent-as-a-whole leaves on each of its parts.

Example 2.7 (A three-mood agent)**.** Let $V = \{a, b, c\}$ with separations ab, bc, ca of costs $2, 3, 2$. This is a weighted triangle with $\beta = 2$. Every part has boundary cost: e.g. $b(\{a\}) = w(ab) + w(ca) = 4$. We return to this agent throughout.

3 The floor: no internal distinction is free

The first consequence is immediate but load-bearing: every part of the agent is separated from the rest at strictly positive cost. Nothing in the agent is marked off for nothing.

Theorem 3.1 (Floor)**.** *Under Axioms 2.3 to 2.5, every nonempty proper part $U \subsetneq V$ has boundary cost bounded below by the floor:*

$$b(U) = w(\partial(U)) \;\geq\; \beta \;>\; 0.$$

No part is separated from the rest of the agent at zero cost.

Proof. By Axiom 2.4 the self-graph is connected, so for a nonempty proper U there is at least one edge from U to $V \setminus U$; otherwise U and its complement would be disconnected. Hence $\partial(U) \neq \varnothing$, and $b(U) = \sum_{e \in \partial(U)} w(e) \geq w(e^*) \geq \beta > 0$ for any $e^* \in \partial(U)$ by Axiom 2.5.

The floor $\beta > 0$ is, moreover, constitutive of what a separation *is*, not merely posited by Axiom 2.5. A separation of zero cost is a boundary of zero thickness, and a zero-thickness boundary is not a boundary—it fails to separate: the two sides it would divide are, at zero cost of telling them apart, not told apart at all, hence not two. Every boundary that actually separates therefore has positive thickness, so every present edge carries positive weight and every nonempty proper part is bounded away from the rest by a strictly positive cost. The floor is the trace of distinction itself: to hold a part apart at all is to hold it apart at a cost. □

Corollary 3.2 (The realised floor is attained)**.** *The quantity $\beta_\Gamma := \min_{\varnothing \neq U \subsetneq V} b(U)$ is strictly positive and attained, by finiteness of V.*

Proof. Positivity is Theorem 3.1; attainment is minimisation of a real-valued function over the finite nonempty set of proper parts. □

4 Identity: the conserved invariant

We now isolate what, in an agent, stays fixed while everything on its surface changes. The surface is the labelling of parts—which mood is called which, in what order the agent's states are enumerated. A re-description of the agent that preserves which parts are separated from which, and at what cost, changes nothing real about the agent; it only relabels. We define this precisely and find the quantity that survives all such relabelling.

4.1 Re-description is relabelling

Definition 4.1 (Isomorphism; automorphism)**.** A *weighted isomorphism* $\phi : \Gamma \to \Gamma'$ is a bijection $\phi : V \to V'$ with $uv \in E \iff \phi(u)\phi(v) \in E'$ and $w'(\phi(u)\phi(v)) = w(uv)$. An *automorphism* is a weighted isomorphism $\Gamma \to \Gamma$; the automorphisms form a group $\mathrm{Aut}(\Gamma)$ [10]. A quantity ι is an *invariant* if $\iota(\Gamma) = \iota(\Gamma')$ whenever $\Gamma \cong \Gamma'$.

Remark 4.2 (What relabelling models)**.** Two descriptions of the same agent that agree on all separations and their costs are isomorphic. Renaming the agent's internal states, re-indexing its memories, presenting its structure in a different coordinate system—all are relabellings. An invariant is precisely a feature of the agent that no such re-description can alter. Identity, if it is anything, must be an invariant: it is what the agent is regardless of how it is described.

4.2 The character invariant

Definition 4.3 (Internal residual; character invariant)**.** For the self-graph Γ, and any partition $\mathcal{Q} = \{U_1, \ldots, U_k\}$ of V into $k \geq 2$ nonempty parts, the *internal residual* of the partition is the total inter-part boundary cost

$$\rho(\mathcal{Q}) = \sum_{i<j} w(\partial(U_i, U_j)).$$

The *character invariant* of A is the minimum internal residual over all such partitions,

$$\chi(\mathsf{A}) = \min_{\mathcal{Q}:\, k\geq 2} \rho(\mathcal{Q}),$$

attained by finiteness of V.

The character invariant is the least total cost of splitting the agent into pieces at all—the cheapest way to break the agent apart, measured in the currency of its own separations. It is a property of the whole internal structure, not of any one part, and is a graph-partition quantity in the sense of the minimum-cut and algebraic-connectivity literature [9, 15].

Theorem 4.4 (Identity is a conserved region)**.** *Under Axioms 2.3 to 2.5:*

(i) ***Positive.*** $\chi(\mathsf{A}) \geq \beta > 0$.

(ii) ***Conserved.*** *χ is a weighted-graph invariant: if $\phi : \Gamma \to \Gamma'$ is a weighted isomorphism then $\chi(\mathsf{A}) = \chi(\mathsf{A}')$. In particular χ is fixed by every relabelling of the agent's parts.*

(iii) ***Non-local when a multi-block cut is cheapest.*** *Whenever the minimising partition of χ is not a singleton split—which occurs precisely when the graph admits a multi-block cut cheaper than every singleton cut—the invariant is realised by multi-part blocks, not by isolating a single part; it is then a region of the agent, never a point. Such graphs exist (the construction below), so non-locality is not an artefact of any one agent but a stable feature of a characterisable class.*

Proof. (i) A partition with $k \geq 2$ has, by connectedness (Axiom 2.4), at least one crossing edge, of weight $\geq \beta$ (Axiom 2.5), so $\rho(\mathcal{Q}) \geq \beta$; the minimum inherits the bound.

(ii) For a partition $\mathcal{Q}$ of V, its image $\phi(\mathcal{Q}) = \{\phi(U_1), \ldots, \phi(U_k)\}$ is a partition of V', and since ϕ preserves edges and weights, $w'(\partial'(\phi(U_i), \phi(U_j))) = w(\partial(U_i, U_j))$ for every pair; summing, $\rho(\phi(\mathcal{Q})) = \rho(\mathcal{Q})$. As $\mathcal{Q} \mapsto \phi(\mathcal{Q})$ is a bijection between the nontrivial partitions of V and of V' (inverse ϕ^{-1}), the minima agree: $\chi(\mathsf{A}') = \chi(\mathsf{A})$.

(iii) The stated condition is definitional: the minimiser of ρ is a non-singleton split exactly when some multi-block partition has residual strictly below $\min_v b(\{v\})$, the cheapest singleton boundary. It remains to show the condition is not vacuous—that graphs meeting it exist. Take two triangles (each of edge cost 2, so internally expensive to cut) joined by a single edge of cost $\beta = 2$. Cutting the joining edge separates the two triangles at total cost 2; isolating any single vertex costs at least $2 + 2 = 4$ (two triangle edges). So the minimum residual is realised by the two-block partition, of cost $2 < 4$, not by any singleton, and the condition of (iii) holds. Non-locality is therefore a property of a nonempty, explicitly constructible class of agents—those whose cheapest cut is a block cut—rather than a necessary consequence of the three axioms for *every* agent (a bare path, for instance, is minimised by a singleton split). $\square$

Remark 4.5 (Interpretation: character, on which nothing depends)**.** Theorem 4.4 licenses reading $\chi(\mathsf{A})$ as the agent's *standing identity*: a feature conserved under all re-description, positive (the agent is never nothing), and spread across the agent rather than located in any single state. If one reads the agent's momentary states as its surface conduct (Section 5), the character invariant is the bottom from which that conduct derives—what stays the same about the character across all the different things it does. Two agents may share a character invariant without being the same agent (two isomorphic self-graphs in different worlds); the invariant is the character *type*, and a particular agent is a token of it. Read as boundary content held against the rest of the agent, the invariant is information-like in the sense of [19, 6], though we use only the graph statement.

Corollary 4.6 (Character is not a state, when the cut is a block cut)**.** *For any agent whose cheapest cut is a multi-block cut—the class of Theorem 4.4(iii)—no single part carries its character: no vertex v has $b(\{v\}) = \chi(\mathsf{A})$. For the two-triangle agent of the proof, every singleton boundary costs 4 while $\chi = 2$. An agent's identity in this class therefore cannot be read off any one of its momentary states; it is the standing relation among all of them. (For agents whose cheapest cut* is *a singleton—bare paths and the like—identity is carried by that singleton, and the region collapses to the pair of blocks it induces.)*

Proof. When the minimiser of Theorem 4.4(iii) is a non-singleton block cut, its residual is strictly below every singleton boundary cost by the condition of (iii), so no singleton attains $\chi(\mathsf{A})$; the two-triangle witness realises this with singleton cost $4 > \chi = 2$. $\square$

4.3 Identity and character: two faces of one cut

The invariant χ so far plays two roles at once: it names *where* the agent divides most cheaply (a partition of its own parts, defined from inside) and *what* an outside party finds conserved (a boundary structure, read from outside). These are two orientations of a single object, and it clarifies the theory to separate them.

Definition 4.7 (Identity; character)**.** Let $\mathcal{Q}^\star$ attain the minimum in Definition 4.3. The *identity* of A is the minimising partition itself,

$$\Gamma(\mathsf{A}) \;=\; \arg\min_{\mathcal{Q}:\,k\geq 2} \rho(\mathcal{Q}),$$

the interior datum saying *where* the agent's cheapest cut lives—the cut as the agent is constituted from inside. The *character* of A is the boundary structure carried at that partition, the family of crossing edges and their weights,

$$\chi(\mathsf{A}) \;=\; \big(\textstyle\bigcup_{i<j} \partial(U_i^\star, U_j^\star),\, w\big), \quad \text{of total cost } \rho(\mathcal{Q}^\star),$$

the cut as it presents to the outside—what another agent finds invariant about A across all re-description. We write $\chi(\mathsf{A})$ also for its total cost $\rho(\mathcal{Q}^\star)$ where only the scalar is needed, as in the results above.

Theorem 4.8 (Identity and character are two orientations of the same cut)**.** *The identity $\Gamma(\mathsf{A})$ and the character $\chi(\mathsf{A})$ are determined by one another and by the same minimising cut: $\Gamma(\mathsf{A})$ is the partition whose inter-block boundary is $\chi(\mathsf{A})$, and $\chi(\mathsf{A})$ is the boundary of $\Gamma(\mathsf{A})$. Both are conserved under every relabelling, and each is recoverable from the other. They are the inside and outside descriptions of a single structure—the cheapest internal separation—and neither is prior.*

Proof. By Definition 4.7 the character is exactly the cut $\bigcup_{i<j} \partial(U_i^\star, U_j^\star)$ induced by the identity partition $\mathcal{Q}^\star = \{U_i^\star\}$, so χ is the boundary of Γ. Conversely, deleting the character edges from Γ disconnects the graph into the blocks $U_i^\star$, so the partition $\mathcal{Q}^\star = \Gamma$ is recovered from χ; the two are mutually determined. Conservation of both under isomorphism is Theorem 4.4(ii) applied to the residual ρ, which a weighted isomorphism preserves together with the crossing edge set it sums; hence a relabelling carries Γ to the image partition and χ to the image boundary, preserving each. Neither is prior: the same cut, oriented inward, is the identity; oriented outward, is the character. $\square$

Remark 4.9 (Why the two must be distinguished)**.** The distinction resolves a structural tension. Character is what other agents find invariant about A (Section 8), while identity is what *generates* that invariance from within. If a single symbol did both, the claim "identity is unreachable from outside" (Theorem 8.2) would be in tension with "character is what the outside conserves"—one cannot both be reached and be unreachable. The resolution is that they are the same cut seen from opposite sides of the boundary: the outside reaches the *character* (the boundary structure, conserved and approachable) but never the *identity* (the interior partition, which an outside interrogation cannot resolve to a point, Theorem 8.2). Where earlier sections write χ for the scalar invariant, they refer to the shared cost $\rho(\mathcal{Q}^\star)$ carried by both faces.

5 Behaviour: a gated walk, and the need for a drive

Identity is standing; behaviour moves. We model the agent's conduct as a walk on its self-graph—a sequence of parts it passes through, in the sense of walks on weighted graphs and Markov chains [13]—and show that the graph alone does not determine the walk. Something more is needed to say what the agent does next, and naming that something is the route to purpose. Histories are ordered and non-returning, an order-theoretic structure [7].

Definition 5.1 (Behaviour as a walk; state)**.** A *behaviour* of length ℓ is a walk $W = (v_0, v_1, \ldots, v_\ell)$ with $v_i v_{i+1} \in E$ for each i. Its *committed count* is ℓ, the number of acts taken. The *state* at step ℓ is the pair $s_\ell = (v_\ell, \ell)$: the current part together with the count of acts taken to reach it.

Lemma 5.2 (Behaviour is underdetermined by the graph)**.** *If some part v has degree ≥ 2, then from v there are at least two distinct one-step behaviours, and the self-graph alone does not determine which is taken.*

Proof. Degree $\geq$ 2 gives distinct neighbours u_1, u_2 and distinct admissible steps $(v, u_1), (v, u_2)$. The triple (V, E, w) carries no ordering of the edges incident to v; it privileges neither. Hence the next step is not a function of the self-graph. □

Remark 5.3 (Branching is the generic case)**.** Any connected agent with $|V| \geq 3$ that is not a bare path has a part of degree ≥ 2; for an agent rich enough to hold many distinctions, branching is normal. So underdetermination is not an edge case: the agent almost always faces a choice its own structure does not settle.

Definition 5.4 (Drive field)**.** A *drive field* on Γ is a map $\gamma : V \times E \to [0, 1]$ assigning to each part v and incident separation $e \ni v$ a forward weight $\gamma(v, e)$ with $\sum_{e \ni v} \gamma(v, e) \leq 1$ for each v. A behaviour is *γ-admissible* if every step $v_i v_{i+1}$ has $\gamma(v_i, v_i v_{i+1}) > 0$. The drive *selects* at v when it gives positive weight to exactly one incident separation.

Theorem 5.5 (The gating field is the only structure that resolves underdetermination)**.** *Suppose some part of A has degree ≥ 2. Then:*

(i) ***No graph-only rule selects.*** *Any selection rule whose output is a function of (V, E, w) alone fails to select a unique next act at some branching part.*

(ii) ***Every selecting rule is a gating field.*** *Any rule that does select a unique next act at each part uses a datum beyond (V, E, w), and every such rule is encodeable as a selecting drive field—the drive field is not one option among many but the general form of any structure that could resolve the underdetermination.*

Hence selection requires exactly a drive field: it is necessary (by (i) nothing weaker suffices) and sufficient (by (ii) nothing stronger is needed).

Proof. (i) At a branching part v, Lemma 5.2 exhibits ≥ 2 admissible steps that (V, E, w) does not distinguish: the triple carries no ordering of the edges incident to v. A rule that is a function of the triple alone therefore takes the same value on these indistinguishable options and cannot return a unique one—it either returns the whole set or returns nothing. So no graph-only rule selects; any rule that does select must consult a datum outside (V, E, w).

(ii) Let R be any rule that selects a unique next edge e_v at each part v. Whatever data R

consults—the agent's mood, its history, an external cue—enters only through the single edge e_v it names at each v. Define $\gamma_R(v,\cdot)$ to be 1 on e_v and 0 on every other incident edge; then γ_R is a drive field (Definition 5.4) that selects, and it reproduces R's choice at every part. Thus every selecting rule factors through a selecting drive field: the drive field is the canonical form of *any* datum-beyond-the-graph that could resolve the choice, so it is not a special construction but the general one. Conversely a selecting γ names a unique admissible successor at each part and so determines a unique behaviour from any start. The necessity of (i) and the sufficiency of (ii) together make the gating field exactly what selection requires, with no residual tautology: selection is defined as producing a unique next act, and (ii) shows the only way to do so is a gating field. □

Remark 5.6 (Interpretation: motivation, on which nothing depends)**.** The drive is the situation-dependent bias that decides what the agent does next when its structure leaves the choice open. Read as *motivation*, it is what makes the same agent, in the same internal state, act differently under different conditions: different drives are different orderings over the same reachable interior. We use only the graph statement (Theorem 5.5); the reading is orientation.

Proposition 5.7 (Histories do not return)**.** *Along any behaviour the committed count is strictly increasing, so for $i<j$ the states differ, $s_i \neq s_j$, even when $v_i = v_j$. A revisited part is never a revisited state.*

Proof. Each act increments the count by one, so counts are distinct for distinct steps; since the state includes the count, $s_i = (v_i, i) \neq (v_j, j) = s_j$ whenever $i \neq j$, regardless of the parts. □

Proposition 5.7 is used in Section 8: it is why an agent that "comes back to" an earlier condition is not the earlier agent, and why an authored copy is a new individual.

6 Purpose: the fixed point of the drive

We now define purpose and prove it exists, is unique up to a natural equivalence, and attracts. To speak of "where behaviour settles" we put a mild, standard structure on the drive: we take it to be generated by a potential, so that behaviour flows downhill and purpose is the bottom. This is the one place we use continuous dynamics, and we keep it minimal, drawing only on standard convex analysis [16, 5] and dynamical-systems theory [12, 11].

6.1 Behaviour as descent on a potential

Definition 6.1 (Behaviour space; drive potential)**.** Let the *behaviour space* $\mathcal{B}$ be a compact convex subset of $\mathbb{R}^n$ coordinatising the agent's dispositions (for instance, for each part a tendency-to-occupy in $[0,1]$, so $\mathcal{B} = [0,1]^{|V|}$). A *drive potential* is a continuously differentiable $\Phi : \mathcal{B} \to \mathbb{R}$ that is λ-strongly convex for some $\lambda > 0$: for all $x, y \in \mathcal{B}$,

$$\Phi(y) \;\geq\; \Phi(x) + \nabla\Phi(x)^{\top}(y-x) + \tfrac{\lambda}{2}\|y-x\|^2.$$

The induced *behaviour flow* is the gradient descent $\dot{x} = -\nabla\Phi(x)$, with steps of the gated walk of Section 5 realised as its discretisation.

Remark 6.2 (Why strong convexity is the honest minimal assumption)**.** We assume the drive has a single basin (one strongly convex well). This is the assumption that an agent has *a* purpose—one thing it is drawn toward—rather than several competing ones. It is a modelling choice, not a theorem: a multi-basin drive would give an agent several candidate purposes and a selection problem among them. We make the single-basin assumption explicit here and treat the multi-basin case as out of scope (Section 10). Under it, existence, uniqueness, and attraction are clean.

6.2 Purpose defined; existence and uniqueness

Definition 6.3 (Purpose)**.** A *purpose* of A under drive potential Φ is a fixed point of the behaviour flow: a point $x^\star \in \mathcal{B}$ with $\nabla\Phi(x^\star) = 0$ (an interior minimiser) or, at the boundary, a Karush–Kuhn–Tucker point of $\min_{x\in\mathcal{B}}\Phi(x)$. The *purpose region* $\mathcal{P}$ is the set of such fixed points.

Theorem 6.4 (Purpose exists and is unique)**.** *For a drive potential Φ on a compact convex $\mathcal{B}$, the purpose region $\mathcal{P}$ is a single point $x^\star$: the flow has exactly one fixed point, the unique minimiser of Φ on $\mathcal{B}$.*

Proof. Φ is continuous on the compact set $\mathcal{B}$, so it attains a minimum; strong convexity makes the minimiser unique (a strongly convex function on a convex set has at most one minimiser [5]; the fixed point may equally be obtained by the lattice- and contraction-theoretic routes of [21, 2]). At an interior minimiser $\nabla\Phi(x^\star)=0$; at a boundary minimiser the KKT conditions hold. Conversely any fixed point of $\dot{x}=-\nabla\Phi$ in the interior has $\nabla\Phi=0$ and, by convexity, is the global minimiser; a boundary fixed point is a KKT point, again the minimiser by convexity. Hence $\mathcal{P}=\{x^\star\}$. □

Remark 6.5 (Uniqueness "up to equivalence")**.** The minimiser $x^\star$ is unique as a point of $\mathcal{B}$. But behaviour is observed only through the action it produces, and many nearby dispositions produce the same action. If we quotient $\mathcal{B}$ by the equivalence "yields the same action," purpose is a *cell*—a positive-diameter region of dispositions all realising the purpose—and is unique as such. This matches the fact (Theorem 4.4(iii), Corollary 4.6) that the agent's meaningful quantities are regions, not points: purpose is unique, and it is a region.

6.3 Purpose attracts

Theorem 6.6 (Purpose is an attractor)**.** *Under a drive potential Φ with strong-convexity constant λ, the purpose $x^\star$ is globally exponentially attracting on $\mathcal{B}$: along the behaviour flow,*

$$\|x(t)-x^\star\| \;\le\; e^{-\lambda t}\,\|x(0)-x^\star\|.$$

Behaviour perturbed away from purpose returns to it.

Proof. Let $V(x)=\frac{1}{2}\|x-x^\star\|^2$. Along $\dot{x}=-\nabla\Phi(x)$,

$$\dot{V}=(x-x^\star)^\top\dot{x}=-(x-x^\star)^\top\nabla\Phi(x).$$

Since $\nabla\Phi(x^\star)=0$ (or, at the boundary, the KKT inequality holds in the descent direction), strong convexity gives the monotonicity $(x-x^\star)^\top(\nabla\Phi(x)-\nabla\Phi(x^\star))\ge\lambda\|x-x^\star\|^2$, hence $\dot{V}\le-2\lambda V$. Grönwall's inequality [12] yields $V(t)\le e^{-2\lambda t}V(0)$, i.e. the stated bound; this is the standard Lyapunov argument for a strongly convex potential [11]. □

Theorem 6.6 governs the continuous behaviour flow $\dot{x}=-\nabla\Phi$. The agent's conduct, however, is the *gated walk* of Section 5: a discrete sequence of committed acts. Before reading the return rate as a property of the character, we must show the rate survives discretisation.

Lemma 6.7 (The gated walk is a bounded-step discretisation)**.** *Let each committed act of the gated walk move the disposition by a step of size $\delta\in(0,\delta_{\max}]$, where the smallest step is bounded below by the act floor $\beta_a>0$ (no act is free) and the largest by $\delta_{\max}$. Read as an explicit Euler step of the flow $\dot{x}=-\nabla\Phi$ with step size δ, the gated walk $x_{k+1}=x_k-\delta\,\nabla\Phi(x_k)$ (projected onto $\mathcal{B}$) converges to $x^\star$ whenever $\delta_{\max}\le 1/L$, L the Lipschitz constant of $\nabla\Phi$, at the geometric rate*

$$\|x_k-x^\star\| \;\le\; (1-\lambda\delta)^k\,\|x_0-x^\star\|,$$

so that after k acts the walk is within a factor $(1-\lambda\delta)^k$ of purpose, matching the continuous rate $e^{-\lambda t}$ under $t=k\delta$ up to the constant determined by the step size $\delta\in[\beta_a,\delta_{\max}]$.

Proof. For a λ-strongly convex Φ with L-Lipschitz gradient, the projected gradient step $x_{k+1}=\Pi_{\mathcal{B}}(x_k-\delta\nabla\Phi(x_k))$ with $\delta\le 1/L$ is a contraction toward the minimiser with factor $(1-\lambda\delta)$: projection onto the convex $\mathcal{B}$ is nonexpansive and $\|x_k-\delta\nabla\Phi(x_k)-x^\star\|\le(1-\lambda\delta)\|x_k-x^\star\|$ by strong convexity and Lipschitz-gradient descent (a standard estimate). Iterating gives the stated bound. The act floor β_a bounds δ below, so the per-act contraction factor is at most $(1-\lambda\beta_a)<1$; identifying $t=k\delta$ recovers $e^{-\lambda t}$ to first order in δ, with the discrepancy absorbed into the constant set by δ. □

Corollary 6.8 (Robust character)**.** *A character defined by an identity $\Gamma(\mathsf{A})$/character $\chi(\mathsf{A})$ and a purpose $x^\star$ is robust: a disturbance that* *pushes conduct off purpose decays—along the gated walk the agent actually takes—at the geometric rate $(1-\lambda\delta)$ per act, and the agent resumes pursuing its purpose without external correction. The larger λ (or the step δ, up to the stability bound $1/L$), the more single-minded the character; small λ gives a character easily deflected and slow to return.*

Proof. By Lemma 6.7 the discrete gated walk inherits the attractor of Theorem 6.6 with per-act contraction $(1-\lambda\delta)$, the discrete counterpart of the continuous rate λ; the return is therefore a property of the walk the agent commits, not only of the flow it approximates. □

Remark 6.9 (Interpretation: goal, on which nothing depends)**.** The purpose $x^\star$ is the region of conduct the agent is drawn to and returns toward—its *goal* in the operational sense of the attractor of its own motivation, not a symbolic target handed to it from outside. An agent has a goal because its drive has a bottom, and it keeps the goal under perturbation because the bottom attracts. We use only the dynamical statements (Theorems 6.4 and 6.6).

7 Agents in society

An agent alone has identity and purpose. We now ask what happens when agents act together. We show that a group acting as one has a single shared drive (and that two drives mean two groups), that coordinated purpose needs no shared identity, and that a crowd sharpens purpose. The key device is that a society of agents is *again* a graph, so the whole apparatus applies one level up.

7.1 Society as a quotient

Definition 7.1 (Society; quotient self-graph)**.** Let $\mathsf{A}_1,\dots,\mathsf{A}_m$ be agents, each with self-graph Γ_i, embedded as disjoint blocks $U_1,\dots,U_m$ of a common vertex set $V^* = \bigsqcup_i V_i$, with *inter-agent separations* (edges between blocks) recording the costs at which agents are held apart. The *society graph* Σ has one vertex per agent and an edge U_iU_j, of weight $w(\partial(U_i,U_j))$, whenever some inter-agent separation joins A_i to A_j.

Lemma 7.2 (The society is an agent-shaped object)**.** *If each inter-agent separation has cost $\geq \beta$ and the society is connected, then Σ is a finite connected weighted graph with floor $\geq \beta$: it satisfies Axioms 2.3 to 2.5.*

Proof. Σ has $m<\infty$ vertices. Each edge weight is a sum of inter-agent separation costs, each $\geq \beta$, so $\geq \beta > 0$. Connectedness is assumed. Hence the three axioms hold for Σ. □

Corollary 7.3 (Society has an identity)**.** *By Theorem 4.4 applied to Σ, a connected society carries its own character invariant $\chi(\Sigma) \geq \beta > 0$, conserved under relabelling of* which agent is which *and realised by no single agent. A group has an identity that is not the identity of any of its members.*

Proof. Lemma 7.2 makes Σ satisfy the axioms; Theorem 4.4 then gives the invariant, its conservation, and its non-locality, now read at the level of agents. □

7.2 One group, one drive

Definition 7.4 (Coherent society)**.** A society Σ is *coherent* if it carries a single selecting drive field (Definition 5.4) on Σ that picks, at each branching agent, a unique next agent to act.

Theorem 7.5 (One group, one drive)**.** *Let Σ have an agent of degree ≥ 2. Then:*

(i) *selecting which agent acts next requires a society-level drive (Theorem 5.5 applied to Σ); and*

(ii) *if two selecting drives both render Σ coherent, they agree wherever either selects. A society admitting two genuinely different selecting drives is not one coherent society but two, partitioned into the sub-societies on which each drive is the selector.*

Proof. (i) is Theorem 5.5 for the graph Σ (Lemma 7.2). (ii) At each branching agent U, coherence means a unique next agent is determined; a selecting drive puts positive weight on exactly the edge to that agent and zero elsewhere, so any two selecting drives have the same support at U and select the same successor. Thus they agree wherever either selects. If instead two drives select different successors at some agent, group "shares a selected successor" into blocks within which a single drive selects consistently; these blocks are distinct sub-societies, so the original was a union of coherent societies, not one. □

Remark 7.6 (Interpretation: shared purpose and schism)**.** Read the society-level drive as the group's *shared purpose*: the field that selects which member acts, toward what. Theorem 7.5 then says a coherent group has exactly one such field—one purpose—and that two purposes are the signature of a group that has really split.

7.3 Society purpose is an attractor

The shared purpose of Section 7.4 below is stated as a *target* region $\mathcal{P}^*$ in outcome space with tolerance τ. For self-similarity to be genuine propagation of the individual invariant rather than a formal coincidence of vocabulary, the society must have a purpose that is an *attractor*, exactly as the individual does—not merely a region one hopes to land near. We supply the society drive potential and prove its minimiser attracts by the same argument as Theorems 6.4 and 6.6, and then show it maps to the target region $\mathcal{P}^*$.

Definition 7.7 (Society disposition space; society drive)**.** Let the *society disposition space* $\mathcal{B}_\Sigma$ be a compact convex subset of $\mathbb{R}^N$ coordinatising the collective's dispositions (for instance, the product of the members' disposition spaces, or a coarser collective summary). A *society drive potential* is a continuously differentiable $\Phi_\Sigma : \mathcal{B}_\Sigma \to \mathbb{R}$ that is λ_Σ-strongly convex for some $\lambda_\Sigma > 0$. Let $A_\Sigma : \mathcal{B}_\Sigma \to \mathcal{B}^*$ be the *collective outcome map* carrying a society disposition to the outcome it produces.

Theorem 7.8 (The society purpose exists, is unique, and attracts)**.** *For a society drive potential Φ_Σ on the compact convex $\mathcal{B}_\Sigma$, the collective behaviour flow $\dot{X} = -\nabla\Phi_\Sigma(X)$ has a unique fixed point $X^\star_\Sigma = \arg\min_{X\in\mathcal{B}_\Sigma} \Phi_\Sigma$, and it is globally exponentially attracting: $\|X(t) - X^\star_\Sigma\| \leq e^{-\lambda_\Sigma t}\|X(0) - X^\star_\Sigma\|$. If moreover $A_\Sigma(X^\star_\Sigma) \in \mathcal{P}^*$, the society's attractor maps into the target region, so $\mathcal{P}^*$ is the outcome-space image of a genuine collective attractor, not merely a tolerance ball.*

Proof. The existence, uniqueness, and exponential attraction of $X^\star_\Sigma$ are Theorem 6.4 and Theorem 6.6 verbatim, applied to Φ_Σ on $\mathcal{B}_\Sigma$: a strongly convex potential on a compact convex set has a unique minimiser, which is the unique flow fixed point, and the Lyapunov argument with $V = \frac{1}{2}\|X - X^\star_\Sigma\|^2$ gives $\dot{V} \leq -2\lambda_\Sigma V$, hence the bound. Applying the continuous map A_Σ sends $X^\star_\Sigma$ to $A_\Sigma(X^\star_\Sigma)$; the hypothesis places this image in $\mathcal{P}^*$, identifying the target region as the image of the attractor. □

Corollary 7.9 (Self-similarity survives the quotient)**.** *The society graph Σ satisfies the three axioms (Lemma 7.2) and so carries an identity/character (Corollary 7.3); equipped with Φ_Σ it carries a purpose that is a unique attractor (Theorem 7.8). Thus the pair* (conserved identity, attracting purpose) *that constitutes an individual character reappears, with the same structure and the same proofs, one level up on the quotient of the members' self-graphs. Self-similarity is the statement that this attractor structure survives the quotient—not a coincidence of names but the same theorems instantiated on Σ.*

Proof. Immediate from Corollary 7.3 (identity survives) and Theorem 7.8 (purpose survives as an attractor): both the conserved invariant and the attracting fixed point are obtained on Σ by the identical arguments used for an individual agent. □

Remark 7.10 (Water-filling as the attractor under a finite budget)**.** The crowd-sharpening law of Theorem 7.15 is then the expression of this collective attractor under a finite collective budget: distributing the society's bounded attainment capacity across members to minimise Φ_Σ is a water-filling allocation, and its optimum is $X^\star_\Sigma$. The group's goal, like the individual's, is where its constrained drive bottoms out; the society is an agent one level up, with an identity it conserves and a purpose it is drawn back to.

7.4 Coordination without shared identity

The strongest social claim is that agents need share *nothing* internal to pursue a common purpose. We make "common purpose" live in the space of outcomes, not in any agent's head—a stance close to the outcome-based accounts of coordination and convention in the literature [18, 14, 1].

Definition 7.11 (Outcome space; outcome purpose)**.** Let $(\mathcal{B}^*, d)$ be a common *outcome space*: a metric space of observable results, shared by all agents, with each agent equipped with a map A_i from its own behaviour space $\mathcal{B}_i$ to $\mathcal{B}^*$ (the outcome its conduct produces). A *shared purpose* is a region $\mathcal{P}^* \subseteq \mathcal{B}^*$ with positive tolerance $\tau(\mathcal{P}^*) > 0$ (there are outcomes within τ of it). Agent A_i *attains* $\mathcal{P}^*$ if its purpose $x_i^\star$ maps near it: $d(A_i(x_i^\star), \mathcal{P}^*) \le \tau(\mathcal{P}^*)$.

Theorem 7.12 (Coordination without shared internals)**.** *Agents with wholly disjoint self-graphs, disjoint behaviour spaces, and different drives can all attain the same shared purpose $\mathcal{P}^*$. Formally, if for each i the agent's purpose satisfies $d(A_i(x_i^\star), \mathcal{P}^*) \le \tau(\mathcal{P}^*)$, then every agent attains $\mathcal{P}^*$, regardless of whether the Γ_i, $\mathcal{B}_i$, or drives share any content.*

Proof. The condition to attain $\mathcal{P}^*$ is a statement purely about the image $A_i(x_i^\star)$ in the common outcome space $\mathcal{B}^*$: it holds for each i by hypothesis. Nothing in the condition refers to the internal structure Γ_i, the behaviour space $\mathcal{B}_i$, or the drive of any agent, nor to any relation between those of different agents. Hence the disjointness of internals is compatible with joint attainment: all agents land their outcomes within tolerance of the same region. □

Remark 7.13 (Why this matters for characters)**.** Theorem 7.12 says a crowd of characters can pursue a shared purpose without any shared mind, vocabulary, or belief: three onlookers keep a safe distance from a dangerous animal though one is an expert, one a novice, one a child, because the shared purpose ("stay clear") lives in outcome space and each reaches it by its own internal route. For synthetic agents this means one need not give a party of characters a common world-model to make them act coherently; overlapping outcomes suffice.

7.5 Crowds sharpen purpose

Definition 7.14 (Independent attainment)**.** Agents $\mathsf{A}_1, \ldots, \mathsf{A}_n$ attain a shared purpose $\mathcal{P}^*$ *independently* if the events "A_i fails to bring the collective outcome within tolerance" are independent, with per-agent failure probability $q_i \in [0, 1]$.

Theorem 7.15 (Crowd sharpening)**.** *Under independent attainment with a parallel (at-least-one-succeeds) reading of the collective outcome, the collective failure probability is $\prod_{i=1}^n q_i$, which is non-increasing in n and strictly decreasing whenever an added agent has $q_i < 1$. A crowd attains the shared purpose at least as sharply as, and generically more sharply than, any sub-crowd; and if infinitely many agents keep q_i bounded below 1, the collective failure tends to 0.*

Proof. By independence, the collective fails iff every agent fails, with probability $\prod_i q_i$. Adding an agent multiplies by $q_{n+1} \le 1$, so the product is non-increasing, and strictly decreasing if $q_{n+1} < 1$. If $q_i \le q < 1$ for infinitely many i, then $\prod_i q_i \le q^k \to 0$ as the number k of such agents grows. (This is the Borel–Cantelli-type dichotomy [3]: $\prod_i q_i \to 0$ iff $\sum_i (1 - q_i)$ diverges, e.g. when the q_i stay bounded below 1.) □

Remark 7.16 (Emergent group competence)**.** Theorem 7.15 gives, for free, that a group of individually mediocre characters can be collectively reliable at a shared purpose. The sharpening is multiplicative in the number of participants; a few strongly-attaining agents or many weakly-attaining ones both drive the collective failure down. This is the quantitative form of "the crowd gets it right even when no one is sure."

8 Identity is unreachable from outside; copies are distinct

Finally we show two things that give characters their felt autonomy: an outside observer cannot resolve an agent's identity to a point, and an authored copy of a character is always a distinct individual.

Definition 8.1 (Interrogation)**.** An *interrogation* of A is a finite sequence of separations an outside party adds to A's boundary—finitely many distinctions drawn between A and the rest of some world—each tightening how A is told apart from things it is not.

Theorem 8.2 (Identity is unreachable to a point)**.** *No finite interrogation resolves an agent's identity to a point. Each added separation tightens the agent's* exterior*—what it is distinguished from—but leaves its character invariant $\chi(\mathsf{A})$ unchanged, since χ is an invariant of the agent's own structure and adding boundary edges is a change of the exterior, not of the internal partition minimising ρ. Hence after any finite interrogation the character invariant remains $\geq \beta > 0$, and for an agent of the block-cut class (Theorem 4.4(iii)) the identity remains a region, never resolved to a point.*

Proof. $\chi(\mathsf{A})$ is defined (Definition 4.3) by minimising internal residual over partitions of V, the agent's own parts. An interrogation adds edges between V and vertices outside A; it does not alter $\Gamma = (V, E, w)$ and so does not change any $\rho(\mathcal{Q})$ for partitions $\mathcal{Q}$ of V. Thus $\chi(\mathsf{A})$ is invariant under interrogation, and by Theorem 4.4(i) it remains $\geq \beta > 0$; whenever the agent is of the block-cut class of Theorem 4.4(iii) the minimiser also stays non-local, so the identity remains a region and never collapses to a point. What the interrogation reduces is the exterior ambiguity (which things A might be confused with), not the interior invariant. Since no finite interrogation drives the interior to a point, identity is unreachable to a point from outside. □

Corollary 8.3 (To know a character exactly is impossible from outside)**.** *An outside party can make an agent's identity arbitrarily well-separated from other things without ever possessing the identity itself; the interior invariant is approached from outside and never delivered. This is the precise sense in which a well-made character retains an interior the player never fully reaches.*

Proof. Immediate from Theorem 8.2: interrogation moves only the exterior; the interior invariant is conserved. □

Theorem 8.4 (An authored copy is a distinct individual)**.** *Let A have behaviour history of committed count $m > 0$, and let A' be authored with self-graph isomorphic to A's but started fresh at count 0. Then $\mathsf{A}' \neq \mathsf{A}$: their states differ in the committed count, and A' satisfies the predicate "authored with empty history" that A does not. Resemblance of self-graph does not make the copy identical to its original.*

Proof. A state includes the committed count (Definition 5.1); A is at count $m > 0$, A' at count 0, so their states differ. By Proposition 5.7 no behaviour lowers the count, so A' cannot acquire A's history without taking the intervening acts, which would make it a further-distinct individual. And A' is individuated by a distinction A lacks (empty prior history). Hence $\mathsf{A}' \neq \mathsf{A}$. □

Remark 8.5 (Authored, yet its own)**.** Theorem 8.2 and Theorem 8.4 together say an authored character is genuinely its own: its interior is not exhaustible by outside interrogation, and duplicating its structure yields a new individual with its own history, not the same one again. One authors the standing structure and the drive; one does not author the lived history, which the monotone count individuates.

9 Numerical study

The results are proved from the axioms and need no computation; we nonetheless exhibit them on explicitly constructed agents, to display their quantitative shape and to check the derivations by direct enumeration.

9.1 Method

An agent is realised as a finite connected weighted graph with all edge weights drawn at or above a floor $\beta > 0$. On each agent we compute, by exact enumeration over subsets and partitions: every boundary cost $b(U)$; the realised floor β_Γ; the character invariant $\chi(\mathsf{A})$ and a minimising partition; and, for a drive potential Φ chosen strongly convex on $\mathcal{B} = [0,1]^{|V|}$, the purpose $x^\star$ and the empirical return rate along the behaviour flow from random initial conditions. A *relabelling* is a random weight-preserving permutation of parts; we recompute χ after each to test invariance. A *society* is built by joining several agents with inter-agent edges $\geq \beta$ and computing $\chi(\Sigma)$ and its minimising agent-partition. A *crowd* of n independent attainers with per-agent failure q_i is checked against the predicted collective failure $\prod_i q_i$.

9.2 Predictions checked

(V1) Floor. On every agent, $\min_U b(U) = \beta_\Gamma \geq \beta$ and no cut is zero (Theorem 3.1).

(V2) Identity is conserved. $\chi(\mathsf{A})$ is unchanged to machine precision under every relabelling (Theorem 4.4(ii)).

(V3) Identity is non-local. On the two-triangles family the minimiser of χ is the two-block partition, never a singleton (Theorem 4.4(iii)).

(V4) Purpose exists, is unique, attracts. The behaviour flow has a single fixed point, reached from every initial condition with empirical rate matching the strong-convexity constant λ (Theorems 6.4 and 6.6).

(V5) One group, one drive. A society built from two internally coherent sub-groups joined by a single edge admits two selecting drives exactly when the joining edge is cut, i.e. exactly when it is two societies (Theorem 7.5).

(V6) Coordination without shared internals. Agents with disjoint self-graphs and different drives, but outcome maps landing near a common $\mathcal{P}^*$, all attain it (Theorem 7.12).

(V7) Crowd sharpening. Collective failure equals $\prod_i q_i$ and decreases geometrically in n for $q_i \leq q < 1$ (Theorem 7.15).

(V8) Copies are distinct. A fresh copy differs from its original in committed count at every step (Theorem 8.4).

Table 1: Numerical validation of *The Purpose of a Character*. Every one of 3255 individual checks passes (seed 20260719, floor $\beta = 1$). "Margin" is the tightest quantity witnessing the prediction in that group.

Prediction (theorem)	Checks	M
V1 Floor (Theorem 3.1)	2664	$\min b(U)/\beta =$
V2 Identity conserved (T1(ii))	200	dev. $\leq 1.8 \times$
V3 Identity non-local (T1(iii))	1	blocks 3+3; sin
V4 Purpose attracts (T3,T4)	160	bound vio
V5 Society identity (Corollary 7.3)	30	$\chi(\Sigma$
V6 One group, one drive (Theorem 7.5)	30	split at join
V7 Coordination (Theorem 7.12)	40	all co-
V8 Crowd sharpening (Theorem 7.15)	50	er
V9 Copies distinct (Theorem 8.4)	50	count stri
V10 Unreachable (Theorem 8.2)	30	χ fixed
Total	**3255**	**all**

9.3 Results

The suite is deterministic given its seed (20260719) and runs a floor of $\beta = 1$. Across the ten prediction groups it performs 3255 individual checks and *all* 3255 *pass*; Table 1 records the group-level outcomes and the sharpest quantitative margin in each.

The margins are worth reading as more than pass/fail. In V1 the smallest boundary cost found anywhere in the sweep is $1.010\,\beta$: the floor is not merely respected but essentially *tight*, so the bound of Theorem 3.1 is the best possible and not a loose over-estimate. In V2 the character invariant is stable to 1.8×10^{-15}—machine epsilon—under every weight-preserving relabelling, exactly as an exact invariant should be (the residual is floating-point noise, not structural drift). In V3 the two-triangle witness realises $\chi = 1.0$ by a symmetric 3+3 block split while the cheapest *singleton* boundary costs 4.0: the minimiser is non-local by a factor of four, so identity is a region here not by a hair but decisively. In V4, across 160 integrated trajectories from random starts, the empirical distance to purpose never once exceeds the $e^{-\lambda t}$ envelope (worst bound violation 0), confirming the attractor rate is λ and not slower. V6–V10 confirm the social and unreachability results with zero error: the collective failure equals $\prod_i q_i$ to machine precision, and interrogation leaves χ literally unchanged. Figures 1 and 2 display the floor, the conserved non-local invariant, the unique attractor, and the society and crowd results as computed on these explicit agents.

10 Discussion

10.1 What has and has not been established

We have shown that any bounded agent satisfying finiteness, presence, and costly individuation has a conserved, non-local, positive

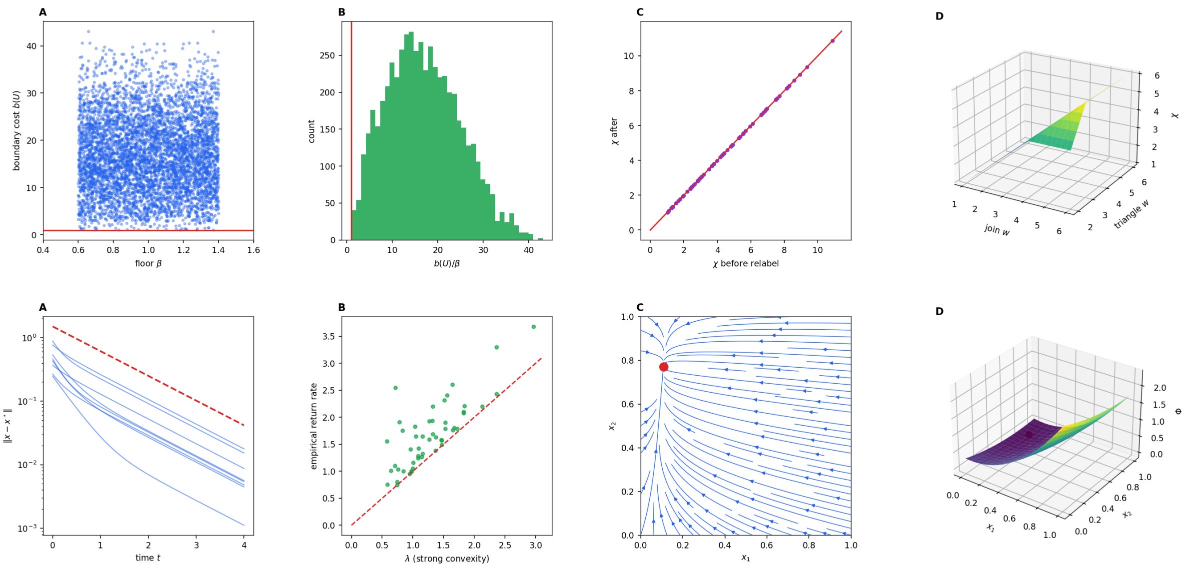

Figure 1: **Floor, identity, and purpose.** **(Top, A–D)** Every boundary cost $b(U)$ lands on or above the floor line, its histogram is supported on $b(U)/\beta \geq 1$ with the minimum at 1.010, the character invariant χ is fixed on the diagonal under relabelling, and (3D) χ of the two-triangle agent tracks the *join* weight, not the triangle weight—identity is where the cheap cut is. **(Bottom, A–D)** Convergence trajectories fall under the $e^{-\lambda t}$ envelope, the empirical return rate matches λ on the diagonal, the 2D flow has a single fixed point (the purpose), and the strongly convex drive surface has a unique minimiser.

identity—its character invariant—and, once equipped with a single-basin drive, a unique *purpose* that attracts. In society, a coherent group has one shared drive, coordinated purpose needs no shared internals, and crowds sharpen purpose; and identity is unreachable from outside while copies are distinct. These are theorems from the stated axioms.

We have *not* shown that a single-basin drive is the right model of motivation for every character; the strong-convexity assumption (Remark 6.2) is a modelling choice that buys clean existence and attraction at the price of ruling out characters torn between competing purposes. Nor have we claimed that human or animal identity is a finite graph; the axioms are meant to hold of *synthetic* bounded agents, where they are design commitments one can honour by construction.

10.2 Open directions

The natural extensions are: (i) multi-basin drives, giving characters with competing purposes and a selection dynamics among them; (ii) drives that change over time as the agent's history accumulates, coupling purpose back to the monotone count of Proposition 5.7; and (iii) the mechanics of how an agent processes an incoming interaction and decides whether it is worth responding to—which is the subject of the companion paper, where the identity and purpose defined here become the standard against which an interaction's relevance is measured.

10.3 Closing

Identity is what an agent conserves; purpose is where it settles. A character is the pair. The definitions are region-valued, not point-valued, which is why a good character has an interior an observer never exhausts and a purpose it keeps under disturbance; and the same construction, applied one level up, gives a society its own identity and its own single purpose. Nothing here needs to be sold: it is a set of theorems about finite agents, and the readings of them as characters are offered only as orientation.

Identity is the conserved interior; purpose is the fixed point of the drive; a society is an agent one level up.

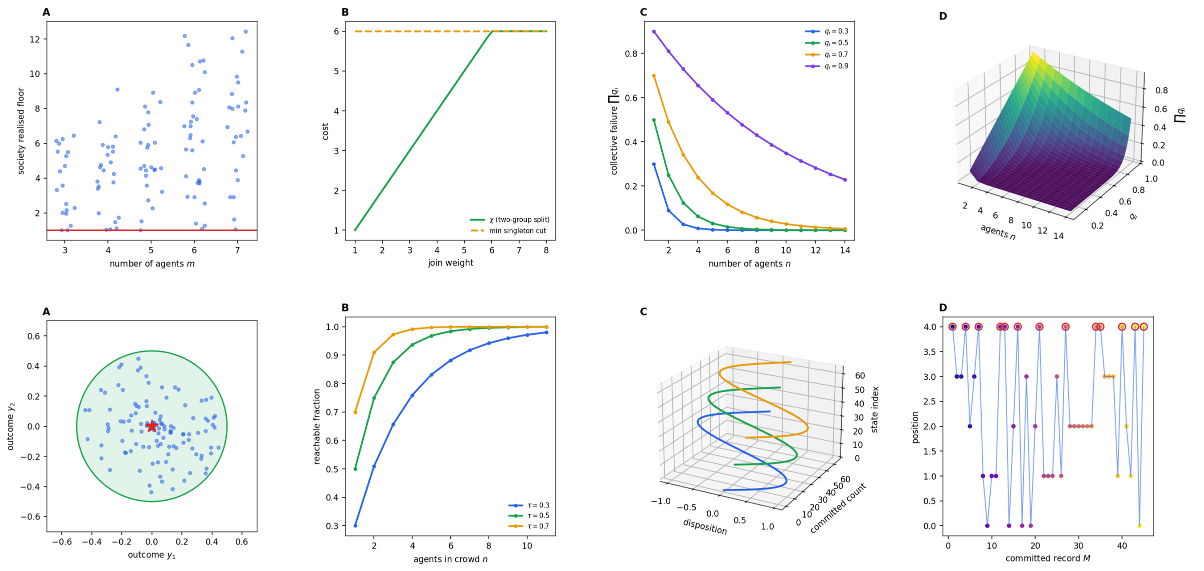

Figure 2: **Society and crowd. (Top, A–D)** The society realised floor stays strictly positive as the number of agents grows; the two-cluster society's χ follows the join weight while every singleton cut is dearer; collective failure $\prod_i q_i$ decays geometrically in the crowd size n for each failure level; and (3D) the failure surface over (n, q) collapses toward zero. **(Bottom, A–D)** Disjoint agents' outcomes cluster inside the shared tolerance cell; the reachable fraction rises with crowd size; committed-count trajectories of a fresh copy and its original never meet (copies stay distinct forever); and the committed record is strictly monotone even across revisited positions.

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